Arjun Raj Singh

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Education

University of Illinois Urbana-Champaign Urbana, IL
Bachelor of Science, Computer Engineering. Expected May 2030

COSMOS, Semiconductor Materials and Device Engineering Santa Cruz, CA
Research and Poster Presentation on EEPROM Architecture and FGMOS Physics. July 2025 – August 2025

Leigh High School San Jose, CA
4.4w, 4.0uw, 36 ACT, NMSC Finalist, AP Scholar w/ Distinction, NHS Expected June 2026
APs: Calc AB (5), BC (5), Chem (5), Phys 1 (5), Mech (5), CSA (5), US Hist (5), Stats, Gov, Lit, Phys 2, E&M, Bio

Projects

8-Bit Microcoded Sequential CISC Breadboard CPU w/ 64KB Addressable Memory Jul 2022 – Present
Architect, Engineer, and Programmer (personal project)

- Designed ISA and data path supporting multi-register operations, stack-based control flow, and memory addressing across 64KB space
- Evolved architecture from a discrete hardware multiplier design, expanding a minimal two-register system into a general-purpose CPU with ALU and microcoded control
- Developed shared memory VGA output terminal, machine code monitor, playable Snake game in custom assembly.
- Constructed ~48-bit horizontal microcoded control unit, 100+ ICs, 40+ breadboards, stable operation at ~1 MHz.
- Utilized oscilloscope analysis to debug propagation timing issues and developed a Python-based assembler toolchain.

Discrete Logic Snake Game Jan 2023 – Jun 2023
Designer, Architect, Engineer (personal project)

- Designed snake game from discrete logic components in Logisim w/o microcontroller or programming
- Runs on 8x8 LED matrix, implementing complete game function directly in digital logic including movement, growth, collisions, and game state control
- Served as precursor to current CPU iteration through analyzing proposed workloads (snake) and useful ALU operations and memory addressing mechanisms

Leadership & Activities

Private Pilot Glider Hollister, CA
PPL-G, Bay Area Soaring Associates Membership Co-Chair *(former student)* Jul 2021 – Present

- 313 flights, 77 hours, 28 PIC hours *ASK-21, SZD-51, DG505, DG1000*

Private Pilot Airplane Hollister, CA
PPL-G, *(former student)* Jun 2024 – Present

- 175 Landings, 73.4 hours, 11 PIC hours *Piper Archer, Cessna 152, Citabria*

Digital Electronics Club *Leigh High School,* San Jose, CA
Founder/President Aug 2024 – Present

- Spearheaded the digital electronics club to share my passion and collaborate with like-minded peers
- Developed presentations and illustrations to guide members with concepts and in completing builds

Skills & Interests

Technical: Computer Architecture, Digital Circuit Design, Assembly, Semiconductor Devices, Python, Java, Arduino
Laboratory: Oscilloscope Analysis, Logisim, 3D Printing, Ceramics